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Pre-logged Biomass and Carbon Stock Density in *Shorea robusta* Gaertn. f. Forests in the Himalayan Foothills of Western Nepal

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ABSTRACT

Background: Forests play a vital role in climate change mitigation by sequestering and storing atmospheric carbon dioxide. However, there is a notable lack of information about the carbon stock density and sequestration capacity of pre-logged Sal (*Shorea robusta*) forests in the Terai region of western Nepal.

Method: This study assessed forest carbon stock in Siddhababa Community Forest (SCF), a pre-logged *Shorea robusta* dominated forest in Kailali District, Nepal. Stratified random sampling was employed using ten 500 m^2 circular plots, covering 0.5% of the forest area. Above ground biomass was estimated using the allometric equations and below ground biomass was calculated using a root-to-shoot ratio of 0.20.

Result: A total of 24 tree species belonging to 19 families were recorded. *Shorea robusta* was the most dominant species with an Importance Value Index (IVI) of 92.06%. Total above ground biomass was 229.51 t ha^{1}, and below ground biomass was 45.90 t ha^{1}. The total carbon stock of the forest was 275.45 tC ha^{1}, which is higher than the national average for mid-hill forests (157.80 tC ha^{1}).

Conclusion: Well-managed community forests in the Terai region have significant carbon stock potential. Regular monitoring and sustainable forest management practices are recommended to enhance carbon stocks and support climate change mitigation efforts.

Key words: pre-logged, biomass, carbon stock, community forest, *Shorea robusta*, Terai

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INTRODUCTION

Forest biomass is not uniformly distributed across the globe; rather, it is shared among different carbon pools, including above ground biomass (tree trunks, branches, leaves), below ground biomass (roots), deadwood, litter, and soil organic carbon. According to the Global Forest Resources Assessment, the total global forest biomass carbon stock is estimated at 296 gigatons, with tropical forests containing approximately 55% of this stock followed by boreal (24%), temperate (14%), and subtropical (7%) forests [1]. Above ground biomass accounts for approximately 70–80% of total forest biomass carbon, while below ground biomass (roots) contributes 20–30% [2]. The sharing of biomass among different forest components has significant implications for carbon accounting and REDD+ initiatives, as each pool responds differently to management interventions and disturbances. Carbon sequestration is the removal of carbon from the atmosphere through storage in the biosphere and is one of the most critical ecosystem services in the context of contemporary climate change. The Intergovernmental Panel on Climate Change reported that forests store approximately 80% of all above ground terrestrial carbon, making forest ecosystems essential for maintaining global carbon balance and mitigating climate change [3]. The Food and Agriculture Organization reported that approximately 31% of the Earth's land surface is covered by forests [4], which are estimated to hold a total carbon content of 289 gigatons in biomass – more than the amount of carbon present in the entire atmosphere. Furthermore, forests store more than 80% of all terrestrial above ground carbon and more than 70% of soil organic carbon [5]. Carbon sequestration varies substantially by forest type, age, structure, and management regime [6]. Reducing Emissions from Deforestation and Forest Degradation (REDD+) has gained major attention in international climate negotiations. Evolving discussions on REDD+ have brought forests to the forefront of both climate change mitigation and adaptation strategies [7]. The Fourth Assessment Report of the IPCC concluded that global climate change is occurring at an alarming rate, with the primary cause being anthropogenic greenhouse gas emissions [8]. Forests play a dual role as both sinks and sources of carbon dioxide, depending on their management and disturbance status [8]. The amount of carbon stored in a forest differs according to spatial and temporal factors such as forest type, size, age, stand structure, associated vegetation, and ecological zonation [9, 10]. Moreover, forests are considered to provide a more cost-effective means of reducing global carbon dioxide emissions than many other sectors. Thus, if incentives can be developed to curb deforestation and forest degradation in tropical countries, forests could have a net positive impact on carbon sequestration and contribute substantially to climate change mitigation. In Nepal, forest covers 59,402 square kilometers, representing 40.4% of the country's total area. Of the 5.96 million hectares of forest area, 2.83 million hectares (47.5% of total forest area) are managed as community forests by more than 4.2 million households. Community forestry is a participatory forest management system initiated in Nepal in the late 1970s. Gilmour and Fisher defined community forestry as the control, protection, and management of forest resources by rural communities for whom trees and forests are an integral part of their farming system [11]. According to the Forest Act of 1993, a community forest is a part or parts of the national forest area handed over to a user group for development, conservation, and utilization for the collective benefit of the community. REDD+ presents an opportunity for developing countries like Nepal to receive payments from developed countries for performance in REDD+ activities, which include reducing emissions from deforestation; reducing emissions from forest degradation; conservation of forest carbon stocks; sustainable management of forests; and enhancement of forest carbon stocks. An operational REDD+ policy requires monitoring of deforestation and forest degradation across entire countries, not only to quantify carbon savings in particular areas but also to account for any compensatory increase in forest biomass loss elsewhere [12]. Despite the clear importance of long-term carbon stock and carbon sequestration in community forests, there is a notable lack of information about the carbon sequestration capacity of pre-logged Sal forests in Kailali District. This study aims to address this gap by providing empirical data on forest biomass and carbon stocks in this pre-logged *Shorea robusta*-dominated forest.

This study aims to assess the pre-logged carbon stock density in Sal dominant community managed forest of Himalayan Foothills of Western Nepal. This study specify the species composition, diversity, and Importance Value Index (IVI) of tree species in Sal dominant Forest. Similary this study has estimated the above ground biomass, below ground biomass, and total carbon stock of the forest and compared the carbon stock with national averages and similar forest types in Nepal.

METHODS

This study was conducted in Siddhababa Community Forest (SCF), located in Godawari Municipality, Ward No. 8, Kailali District, Nepal. The forest was officially recognized as a community forest in 2053 BS (1996 AD) and is named after the Siddhababa Temple located within the forest. The total area of the community forest is 107.014 hectares, which includes roads, lakes, and land-use areas, leaving 88.86 hectares as productive forest area.

The forest is bordered by Ghudsawar to the east, west, and south; Jonapur to the east; and Fakalpur to the north and west. The forest is divided into three blocks: Block A (60.62 ha), Block B (15.13 ha), and Block C (13.12 ha), with a total productive area of 88.88 hectares (Figure 1).

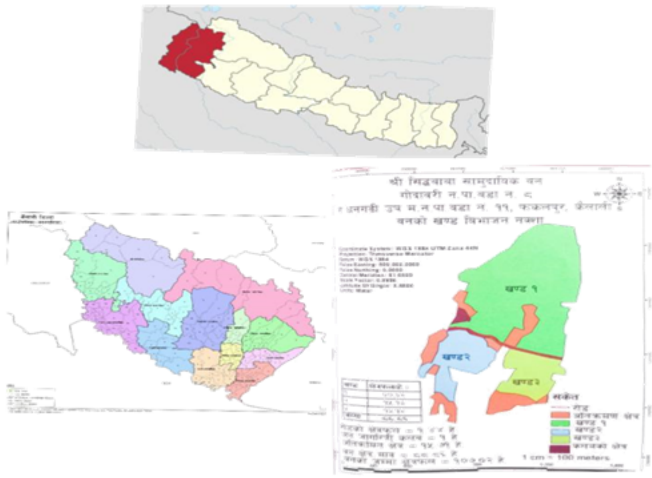

Figure 1. Map of the study area.

The study was conducted in January 2022. A stratified random sampling method was used to locate sampling plots. The study site was selected based on the presence of diverse tree species and sufficient wood biomass, despite its relatively small area of 107.4 hectares.

The forest is dominated by *Shorea robusta*, which is known as a significant carbon-sequestering species in subtropical forests. Twenty circular plots of 500 m^2 were established, covering approximately 0.5% of the forest area. Quadrats were delineated using iron pegs and rope.

All tree species present within the quadrats were recorded. Diameter at breast height (DBH), measured at 137 cm above ground level, was recorded for all individual trees with DBH greater than 10 cm using a DBH tape. Tree height was measured using a clinometer, and the observer's eye height was measured using a measuring tape.

The geographic coordinates (latitude, longitude, and elevation) of each sampling plot were recorded at the plot center using a Global Positioning System (GPS). Slope and aspect were measured using clinometers. Tree height above the observer's eye level was calculated using trigonometric formulas following Thapa Magar [13].

Plant specimens were collected, tagged, and pressed using newspapers and a herbarium press in the field. Local names of most specimens were recorded through consultation with local villagers. Specimens were identified at the Tribhuvan University Central Herbarium (TUCH) and the National Herbarium and Plant Laboratories (KATH). The following vegetation parameters were calculated for each tree species.

Density and Relative Density

Density (per hectare) represents the number of individual trees per unit area and indicates the numerical strength of a species in a community.

$$\text{Density} \;=\frac{\text{Total number of individuals of a species}}{\text{Total number of plots} \times \text{area of each plot (ha)}} \tag{1}$$

$$\text{Relative Density (\%)} = \frac{\text{Density of a species}}{\text{Total density of all species}} \times 100 \tag{2}$$

Frequency and Relative Frequency

Frequency indicates the dispersion of a species within the community and is expressed as the percentage of sampling units in which a particular species occurs.

$$\text{Frequency (\%)} = \frac{\text{Number of plots in which species occurred}}{\text{Total number of plots sampled}} \times 100 \tag{3}$$

$$\text{Relative Frequency (\%)} = \frac{\text{Frequency of a species}}{\text{Total frequency of all species}} \times 100 \tag{4}$$

Coverage and Relative Coverage

Coverage refers to the area of ground occupied by the above ground parts of plants.

$$\text{Coverage} = \frac{\text{Total number of individuals of the species}}{\text{Total number of quadrats in which the species occurred}} \tag{5}$$

$$\text{Relative Coverage (\%)} = \frac{\text{Coverage of a species}}{\text{Total coverage of all species}} \times 100 \tag{6}$$

Important Value Index (IVI)

The Important Value Index (IVI) expresses the ecological dominance of a species as a single value obtained by summing relative density, relative frequency, and relative coverage.

$$\text{IVI} = \text{Relative Density} + \text{Relative Frequency} + \text{Relative Coverage} \tag{7}$$

Above Ground Biomass Estimation

Biomass estimation was based on wood-specific gravity values obtained from the Global Wood Density Database. Above ground biomass was estimated using the allometric equation developed by Chave et al. [14]:

$$AGTB = 0.0509 \times \rho \times D^2 \times H \tag{8}$$

where,

- $AGTB$ = Above ground tree biomass (kg),
- ρ = Wood-specific gravity (g cm^{-3}),
- D = Diameter at breast height (cm),
- H = Tree height (m).

After summing the biomass values of all trees within a sampling plot and dividing by the plot area (500 m^2), biomass stock was obtained in kg m^{-2}. The value was converted into tonnes per hectare (t ha^{-1}) by multiplying by 10.

Biomass stock was converted into carbon stock using the IPCC (2006) default carbon fraction of 0.47.

Below Ground Biomass Estimation

Methods for estimating below ground biomass are not fully standardized (IPCC, 2006). Since destructive excavation was not feasible for this study, a conservative root-to-shoot ratio of 0.20 was applied, as recommended by IPCC (2006) and MacDicken (1997), which is suitable for tropical moist deciduous and subtropical humid forests.

$$BGB = AGTB \times 0.20 \tag{9}$$

Total Carbon Stock

Total carbon stock was calculated as the sum of above ground and below ground carbon stocks:

$$\text{Total Carbon (tC ha}^{-1}\text{)} = \text{Above ground Carbon} + \text{Below ground Carbon} \tag{10}$$

RESULTS

Forest Species Composition

Siddhababa Community Forest (SCF) comprises 24 tree species belonging to 19 different families. The families

Fabaceae and Myrtaceae are represented by the highest number of species, with three species each. Other families, such as Combretaceae, Rubiaceae, Bixaceae, Sapotaceae, Malvaceae, Rutaceae, Meliaceae, Rhamnaceae, Poaceae, Moraceae, Anacardiaceae, Phyllanthaceae, and Verbenaceae, are represented by one or two species.

Although some families are represented by relatively few species, the forest is predominantly dominated by *Shorea robusta* (family Dipterocarpaceae) and *Terminalia elliptica* (family Combretaceae). Together, the families Dipterocarpaceae and Combretaceae account for approximately 70% of the total tree population in the community forest.

Importance Value Index (IVI)

Shorea robusta was found to be the most dominant and ecologically important tree species in SCF, with an IVI of 92.06. *Terminalia elliptica* had the second-highest IVI value of 37.95 (Figure 2). These findings confirm that SCF is a *Shorea robusta*-dominated forest, which is characteristic of the Terai region of Nepal.

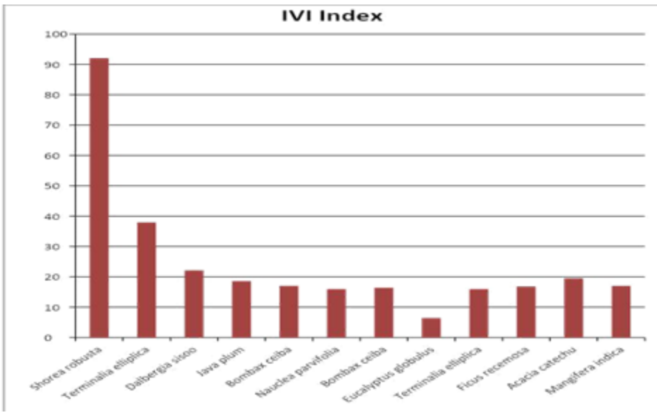

Figure 2. Important Value Index (IVI) of different tree species in SCF.

Above Ground Biomass

The highest above ground biomass values were recorded for *Shorea robusta* and *Terminalia elliptica*, with biomass stocks of 150.10 t ha^{-1} and 145.78 t ha^{-1}, respectively (Figure 3). These two species constitute the majority of the forest biomass due to their dominance in both abundance and tree size.

Carbon Stock in Different Tree Species

Carbon stock varied among tree species owing to differences in their frequency, density, diameter at breast height (DBH), and tree height (Figure 4).

The total carbon stock in the above ground biomass of SCF was estimated to be 229.51 tC ha^{-1}, whereas the below ground carbon stock was estimated at 45.90 tC ha^{-1}. The overall total carbon stock of the community forest was found to be 275.41 tC ha^{-1} (Table 1).

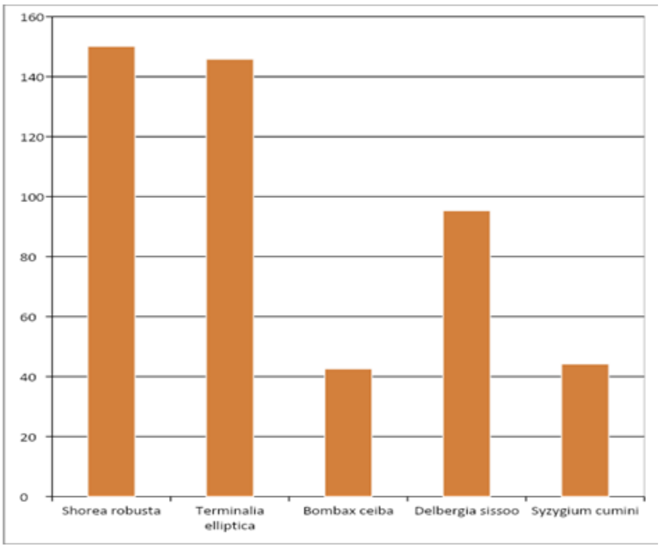

Figure 3. Average above ground biomass of the five major tree species in SCF.

Table 1. Carbon stock distribution in Siddhababa Community Forest.

Carbon Pool	**Carbon Stock (tC ha^{-1})**	**Percentage (%)**
Above Ground	229.51	83.32
Below Ground	45.90	16.68
Total	275.41	100.00

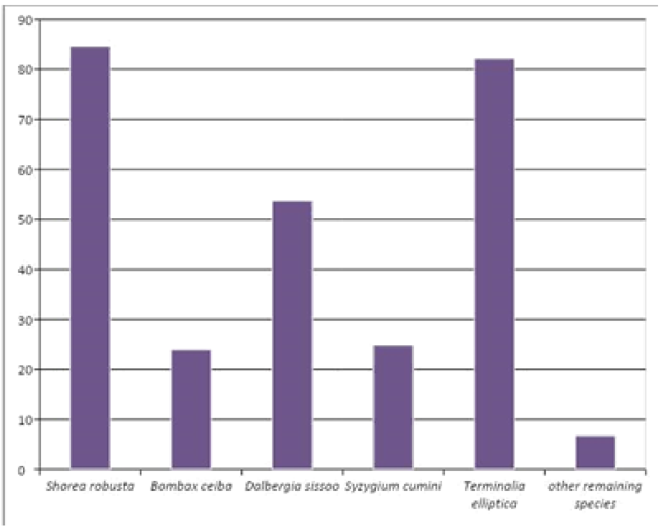

Figure 4. Carbon stock in different tree species in SCF.

DISCUSSION

The total carbon stock of Siddhababa Community Forest (275.45 tC ha^{-1}) is substantially higher than the national average for mid-hill forests (157.80 tC ha^{-1}) reported by the Forest Resource Assessment (FRA, 2014). This difference can be attributed to several factors, including the forest's location in the productive Terai region, the dominance of *Shorea robusta* (a high-biomass species), and the protection afforded by community forest management.

The carbon stock in SCF is comparable to that of dense strata in REDD+ pilot watersheds in Nepal. For example, Kayarkhola watershed had a carbon stock of 296.44 tC ha^{-1} in dense strata, while Charnawati and Ludhikhola watersheds had 228.56 tC ha^{-1} and 216.26 tC ha^{-1}, respectively (Anup et al., 2013). The value of 275.45 tC ha^{-1} found in SCF falls within this range, confirming that community-managed *Shorea robusta* forests in the Terai have carbon sequestration potential comparable to the best-performing REDD+ pilot sites.

In contrast, Thagunna (2009) reported an above ground biomass density of only 26.03 t ha^{-1} for Bailbandh Community Forest in Kanchanpur, which has a similar forest type but a different management history. The substantially higher biomass in SCF likely reflects differences in forest age, disturbance history, and management effectiveness. Similarly, the ICIMOD Knowledge Park, a strictly protected forest in a subtropical zone, had a total forest biomass of 186.43 t ha^{-1} in 2014 (ICIMOD, 2016), which is lower than the 229.51 t ha^{-1} above ground biomass found in SCF. This difference may be due to the different ecological conditions between the mid-hills (Godavari) and the Terai (Kailali), as the Terai region generally supports higher biomass production due to more favorable growing conditions.

The DBH class distribution in SCF, with 34% of trees in the 10–20 cm class and 20% in the less than 10 cm class, indicates a young, regenerating forest with high future carbon sequestration potential. This "pole stage" structure is characteristic of well-managed community forests where selective harvesting has occurred but regeneration is actively protected. As these young trees mature, biomass accumulation rates are expected to remain high for several decades.

The dominance of *Shorea robusta* (IVI = 92.06) is consistent with the ecology of the Terai region, where Sal forests are the climax vegetation type. The high IVI value reflects the species' superior competitive ability, high frequency across sampling plots, and large basal area contribution. The presence of 23 other tree species indicates that, despite Sal dominance, the forest maintains significant biodiversity, which is important for ecosystem resilience and ecological functioning.

CONCLUSIONS

This study provides the first quantitative assessment of forest biomass and carbon stock in Siddhababa Community Forest (SCF), Kailali District, Nepal. The forest contains 24 tree species belonging to 19 families, with *Shorea robusta* identified as the most dominant species (IVI = 92.06). The DBH class distribution (34

The total carbon stock of the forest was estimated at 275.45 tC ha^{-1}, which is higher than the national average for mid-hill forests and comparable to the dense strata reported from REDD+ pilot sites in Nepal. These findings demonstrate that well-managed community forests in the Terai region possess significant carbon sequestration potential and can contribute meaningfully to climate change mitigation efforts.

Regular monitoring using permanent sample plots, the adoption of sustainable forest management practices, and the protection of natural regeneration are recommended to maintain and enhance carbon stocks in SCF and similar community forests of Nepal.

Based on the findings of this study, the following recommendations are proposed:

1. **Regular Monitoring:** Permanent sample plots should be established and monitored periodically (every five years) to track changes in carbon stocks and assess the effectiveness of forest management interventions.

2. **Sustainable Harvesting Practices:** Selective harvesting should be carried out in accordance with community forest operational plans that maintain forest structure and promote carbon stock regeneration. High-biomass species such as *Shorea robusta* and *Terminalia elliptica* should be prioritized for retention.

3. **Protection of Regeneration:** Natural regeneration should be protected through controlled grazing and effective fire management practices to ensure the continuous recruitment of young trees, thereby sustaining future carbon sequestration.

4. **Biodiversity Conservation:** Despite the dominance of Sal (*Shorea robusta*), the remaining 23 tree species should be conserved to enhance ecosystem resilience and adaptability under changing climatic conditions.

5. **REDD+ Readiness:** The baseline carbon stock data generated from SCF can contribute to the development of sub-national reference levels for REDD+ initiatives in the western Terai region of Nepal, potentially enabling performance-based payments for community forest user groups.

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